



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Luis Gerardo Reyes Sanchez
May 19th, 2026

GitHub Repository:
<https://github.com/lgerardoreyes14/testrepo>



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

This project analyzes SpaceX Falcon 9 launch data to identify the factors associated with successful first-stage landings.

The project included data collection from the SpaceX API, web scraping, data wrangling, exploratory data analysis, geospatial visualization, interactive dashboarding, and predictive modeling.

Key findings:

- Launch success rates improved over time.
- Payload mass, orbit, launch site, and booster version were relevant factors associated with landing outcomes.
- Exploratory analysis showed differences in success rates across launch sites and mission characteristics.
- Predictive models were trained to classify whether the Falcon 9 first stage would land successfully.
- The best-performing model was selected based on evaluation metrics and confusion matrix results.

Introduction

SpaceX has reduced launch costs by reusing the Falcon 9 first stage. A Falcon 9 launch is advertised at a much lower cost than many competitors, partly because the first stage can be recovered and reused.

The project frames the business problem around determining whether the first stage will land successfully, since this helps estimate launch cost and competitiveness.

- **Business Question:**
Can we predict whether the Falcon 9 first stage will land successfully using historical launch data?
- **Objective:**
Build a data science workflow that collects, cleans, explores, visualizes, and models SpaceX launch data to predict landing success.

Section 1

Methodology

Methodology

This capstone project follows an end-to-end data science workflow to analyze and predict the success of SpaceX Falcon 9 first-stage landings. The methodology includes:

1. Data Collection

- Collected launch data from the SpaceX API.
- Enriched the dataset through web scraping of historical launch records.

2. Data Wrangling

- Cleaned and transformed raw launch data.
- Created the target variable representing landing success.
- Prepared structured datasets for analysis, visualization, and modeling.

3. Exploratory Data Analysis

- Used SQL queries to summarize launch sites, payloads, orbit types, success rates, rankings, and time trends.
- Built visualizations to identify patterns between launch outcomes and mission features.

Methodology

4. Interactive Visual Analytics

- Used Folium to map launch sites, launch outcomes, and proximity to key geographic features.
- Used Plotly Dash to build an interactive dashboard with launch-site filtering, success pie charts, and payload-versus-outcome scatter plots.

5. Predictive Analysis

- Built classification models to predict whether the Falcon 9 first stage would land successfully.
- Tuned and evaluated models using classification metrics and confusion matrices.
- Compared model performance to select the most effective predictive approach.

The final goal is to understand the operational factors associated with successful landings and build a model capable of predicting landing success using historical SpaceX launch data.

Data Collection

The project used two complementary data collection methods to build the SpaceX Falcon 9 launch dataset.

1. SpaceX API Data Collection

Launch data was collected from the SpaceX API to retrieve structured mission information, including launch dates, rocket details, payload data, orbit information, launch sites, and landing outcomes.

Process:

SpaceX API → JSON response → Data extraction → DataFrame creation → Clean launch dataset

2. Web Scraping Data Collection

Additional launch records were collected through web scraping to complement the API dataset and validate mission information.

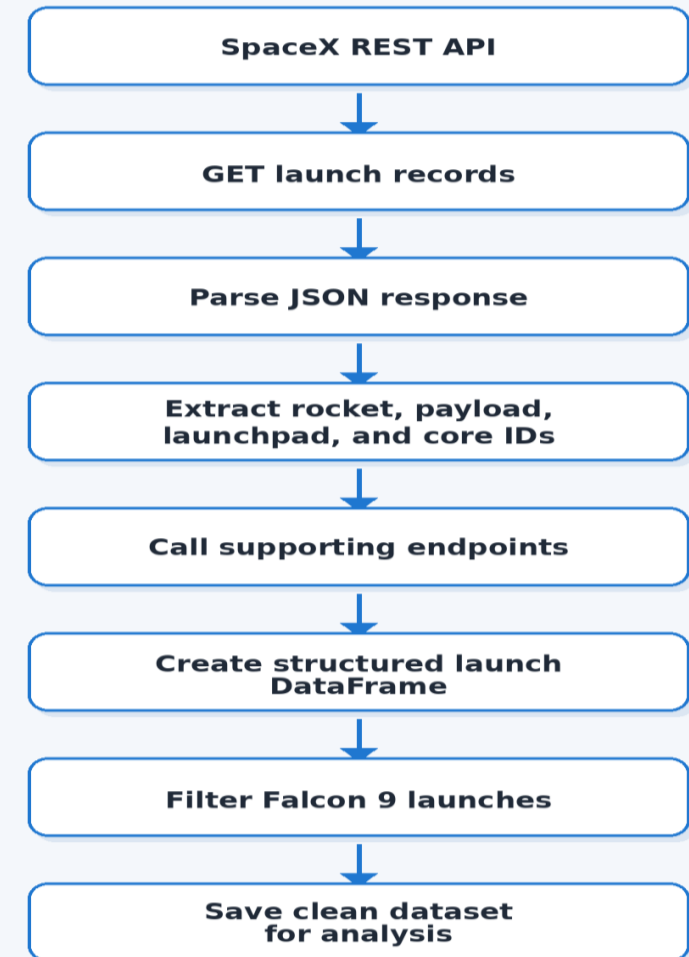
Process:

Web page → HTML table → BeautifulSoup parsing → Table extraction → Clean tabular dataset

Data Collection – SpaceX API

Key API collection steps:

- Requested historical launch records from the SpaceX API.
- Retrieved launch-level information such as flight number, launch date, rocket ID, payloads, launchpad, and cores.
- Used related API endpoints to enrich the dataset with:
 - Booster version
 - Payload mass
 - Orbit
 - Launch site
 - Longitude and latitude
 - Landing outcome
- Converted nested JSON data into a structured pandas DataFrame.
- Filtered the dataset to Falcon 9 launches.
- Prepared the collected data for later data wrangling, EDA, dashboarding, and machine learning.



GitHub URL:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/01-data-collection-api.ipynb>

Data Collection - Scraping

Key scraping steps:

- Requested the Wikipedia page containing Falcon 9 and Falcon Heavy launch records.
- Parsed the page HTML using BeautifulSoup.
- Located the launch records table.
- Extracted column headers and table row values.
- Cleaned HTML elements such as links, superscripts, and line breaks.
- Converted the extracted records into a pandas DataFrame.
- Prepared the scraped dataset for data wrangling, EDA, visualization, and modeling.

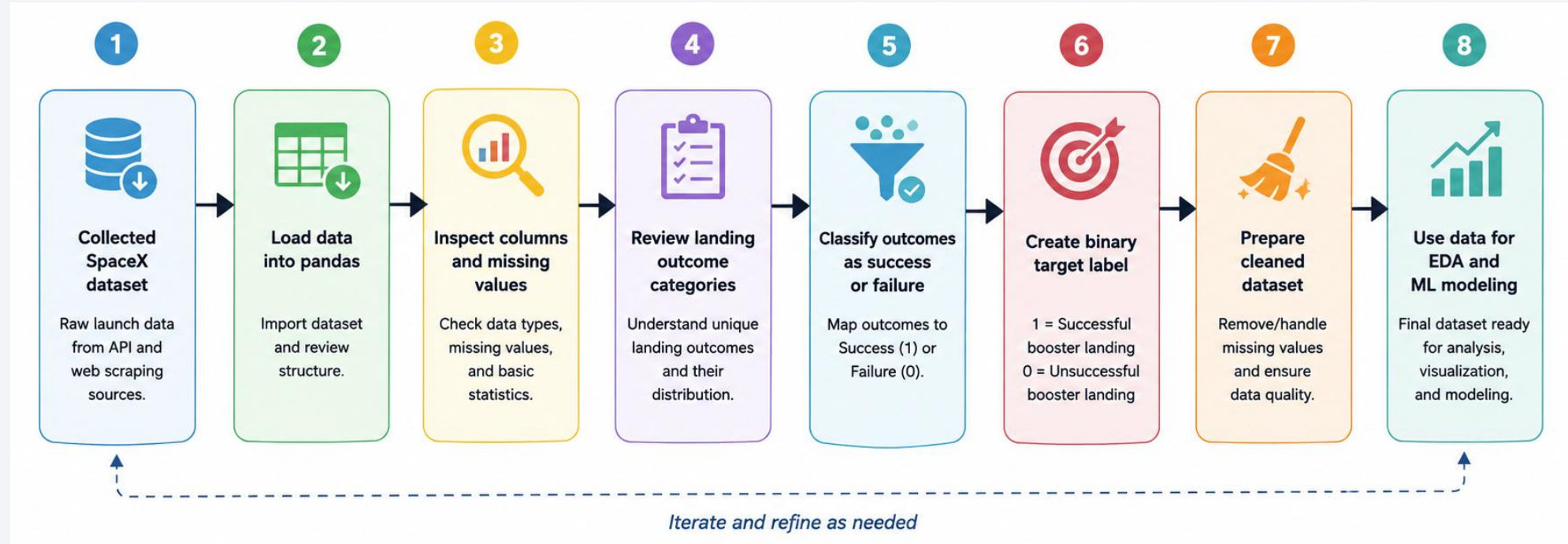


GitHub URL:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/02-webscraping.ipynb>

Data Wrangling

Objective: Prepare the SpaceX launch dataset for exploratory analysis and supervised machine learning.



GitHub URL:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/03-data-wrangling.ipynb>

EDA with Data Visualization

Objective:

Identify visual patterns associated with Falcon 9 landing success.

Charts used:

- Scatter plots: payload mass and flight number vs. launch site/orbit
- Bar charts: success rate by orbit
- Trend plots: yearly success rate over time

Purpose:

Compare success patterns across launch sites, payload mass, orbit type, and launch year to support feature selection and predictive modeling.

EDA with SQL

Objective:

Analyze SpaceX launch records using SQL to summarize launch sites, payloads, boosters, landing outcomes, and time trends.

Queries performed:

- Unique launch sites
- Launch records by site prefix
- Payload mass summaries
- Booster and payload filtering
- Landing outcome counts and rankings
- Success/failure analysis by date range
- Time-based launch outcome analysis

GitHub URL:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/04-eda-sql.ipynb>

Build an Interactive Map with Folium

Objective:

Build an interactive Folium map to explore SpaceX launch site locations and nearby geographic features.

Map objects:

- Launch site markers with popups
- Circle markers for successful and failed launches
- Distance markers for coastline, highway, railway, and city proximity
- Lines connecting launch sites to nearby reference points

Purpose:

Visualize launch site geography, compare launch outcomes by location, and evaluate proximity to operationally relevant infrastructure.

GitHub URL:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/06-launch-site-map-folium.ipynb>

Build a Dashboard with Plotly Dash

Objective: Create an interactive dashboard to explore SpaceX launch success patterns by launch site, payload mass, and booster version.

Dashboard components and interactions

Launch site dropdown

- Allows users to select all launch sites or filter to a specific site.

Success pie chart

- For all sites, shows total successful launches by launch site.
- For a selected site, shows success versus failure distribution.

Payload range slider

- Allows users to filter launches by payload mass.
- Helps isolate launch outcomes for lighter or heavier payloads.

Payload vs. launch outcome scatter plot:

- Shows the relationship between payload mass and launch success.
- Uses booster version category as the color dimension.

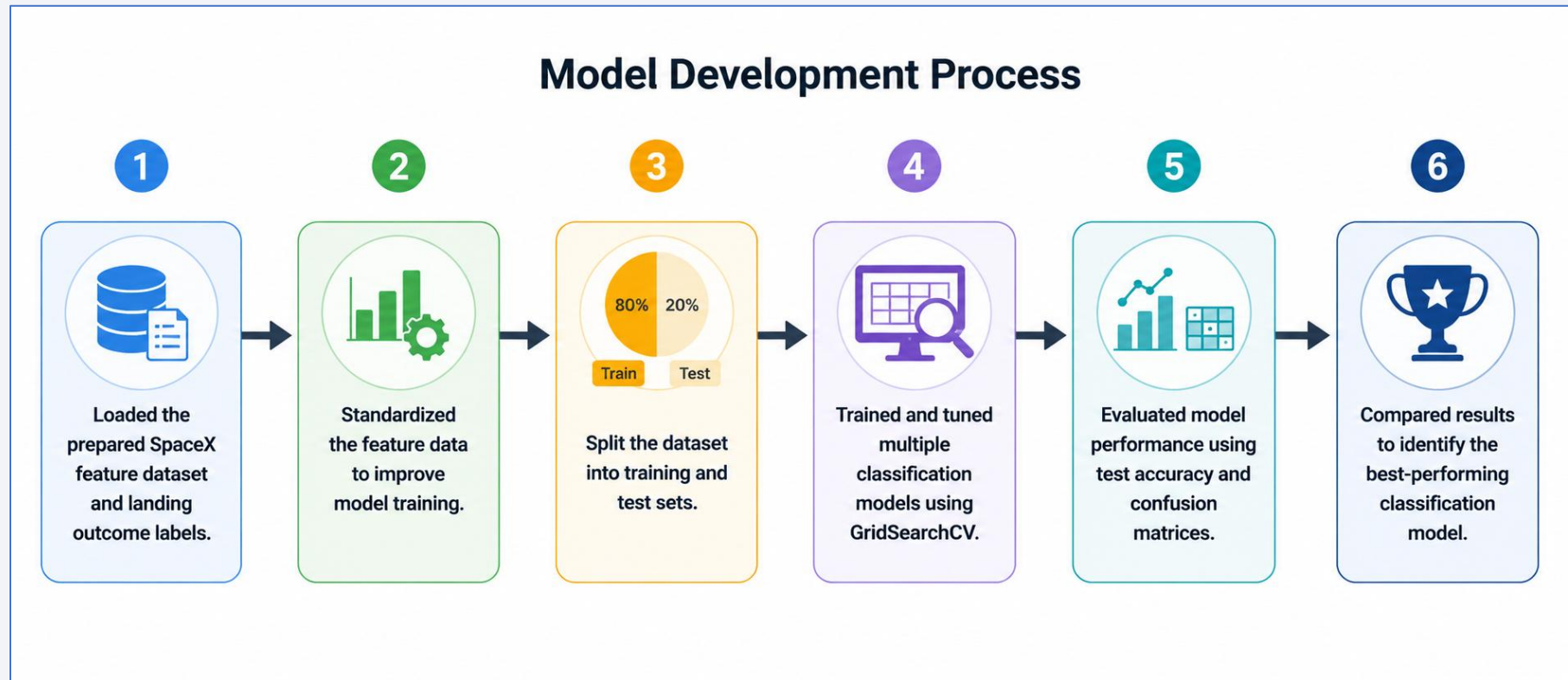
GitHub URL:

https://github.com/lgerardoreyes14/testrepo/blob/main/apps/spacex_dash/app.py

Predictive Analysis (Classification)

Objective:

Build and evaluate classification models to predict whether the Falcon 9 first stage will land successfully.

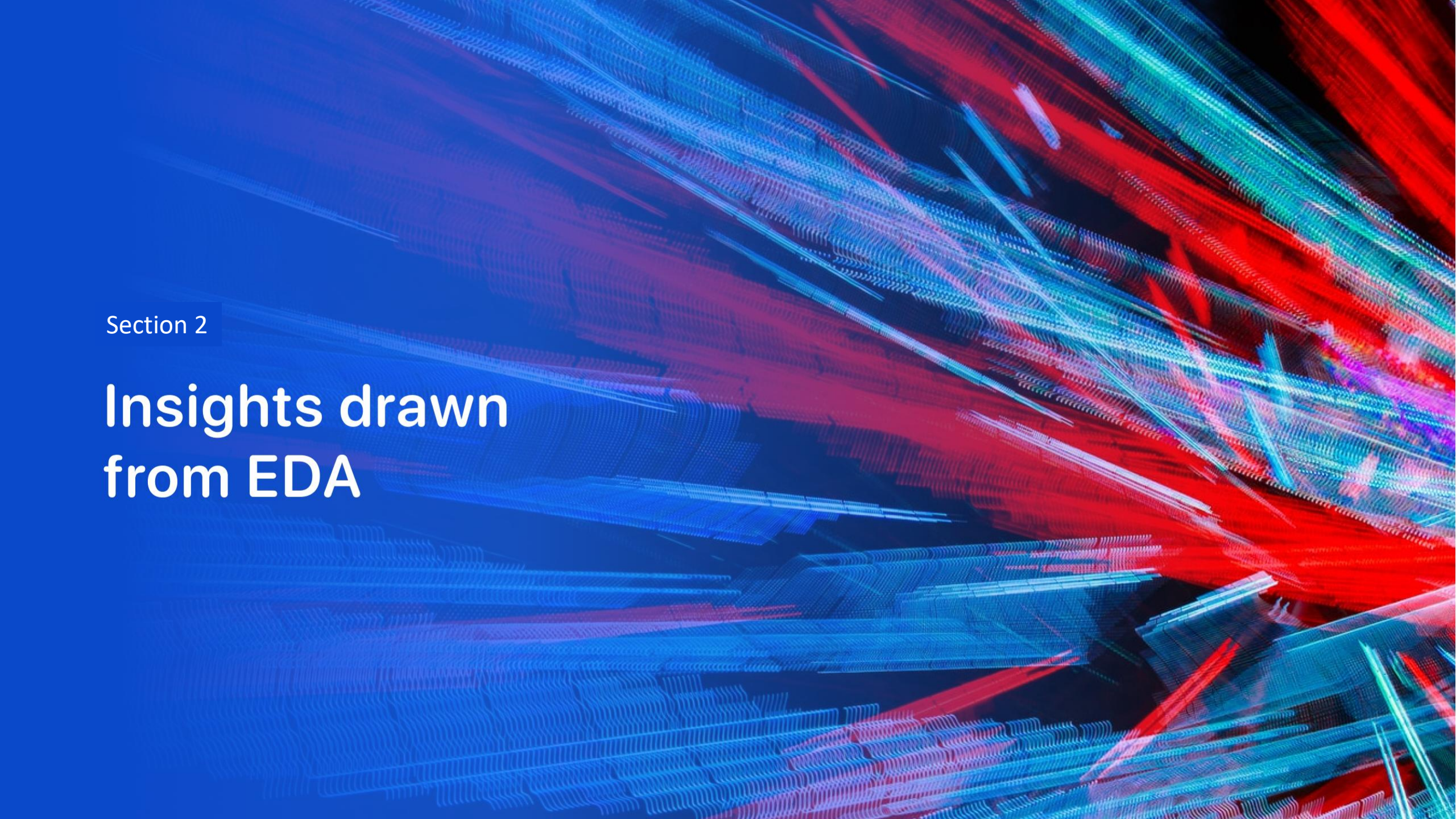


Results

This section presents the main findings from the SpaceX Falcon 9 landing prediction project.

Results are organized into five areas:

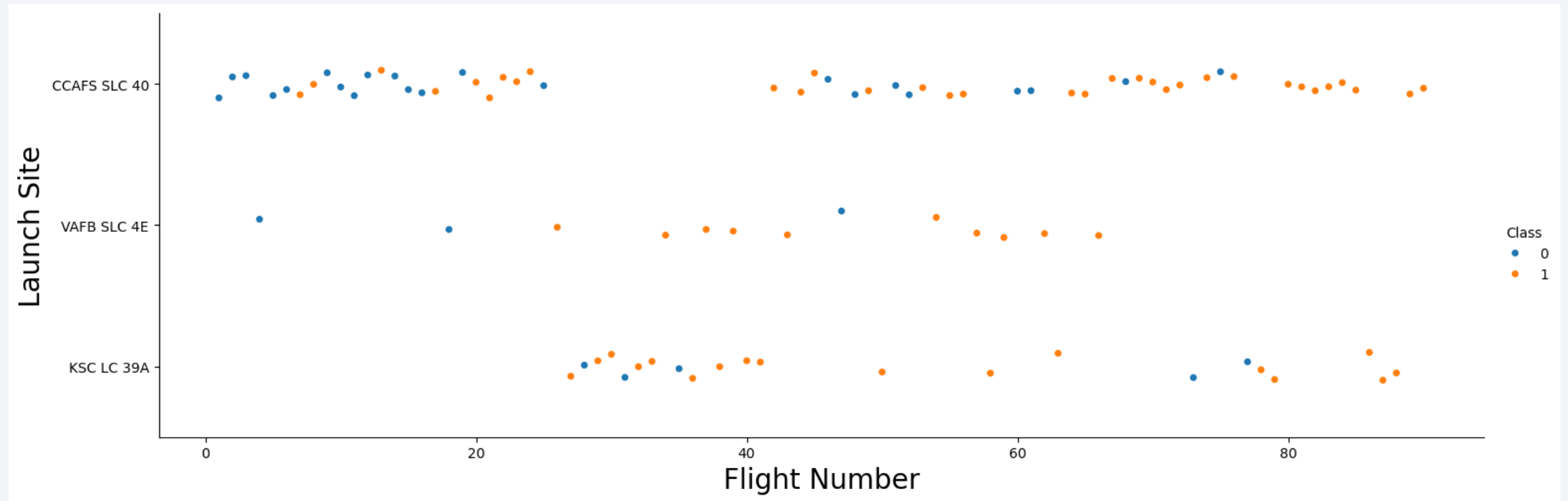
- **EDA visualizations:** patterns across launch sites, payload mass, orbit type, and yearly success trends.
- **SQL analysis:** launch site summaries, payload queries, landing outcome rankings, and time-based results.
- **Folium maps:** launch site locations, outcome markers, and proximity analysis.
- **Plotly Dash dashboard:** interactive success charts and payload-outcome exploration.
- **Predictive modeling:** classification model comparison, best model selection, and confusion matrix interpretation.

The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and teal on the right. These streaks have a textured, almost woven appearance. Overlaid on this pattern is a faint, light blue grid that recedes into the distance, creating a sense of depth and perspective.

Section 2

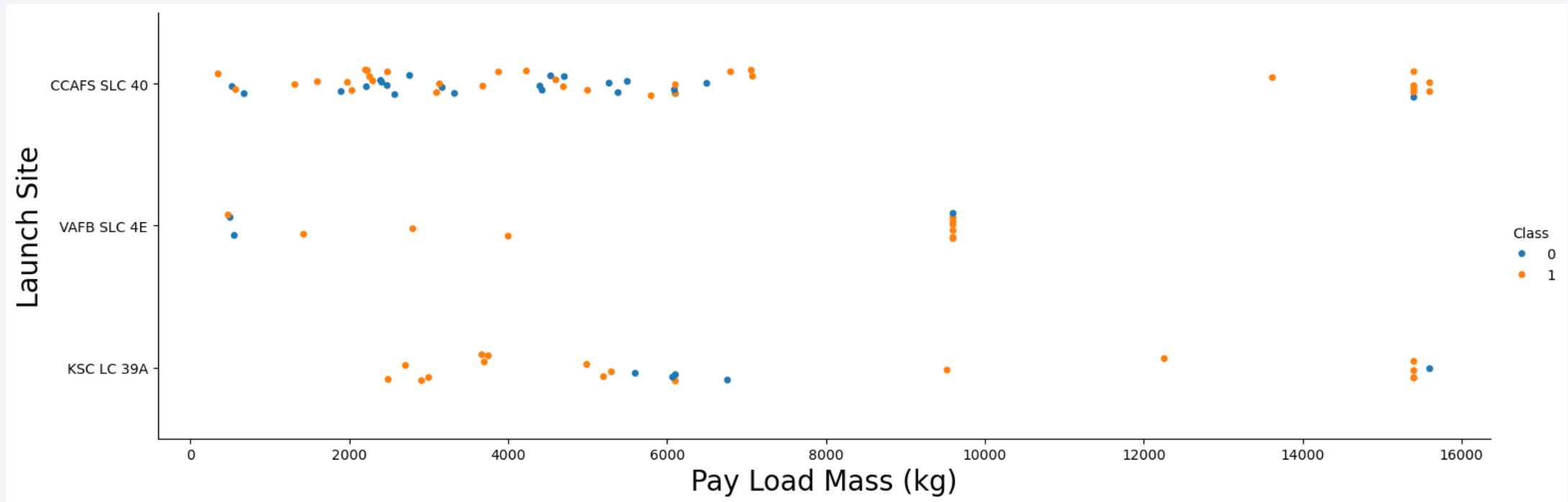
Insights drawn from EDA

Flight Number vs. Launch Site



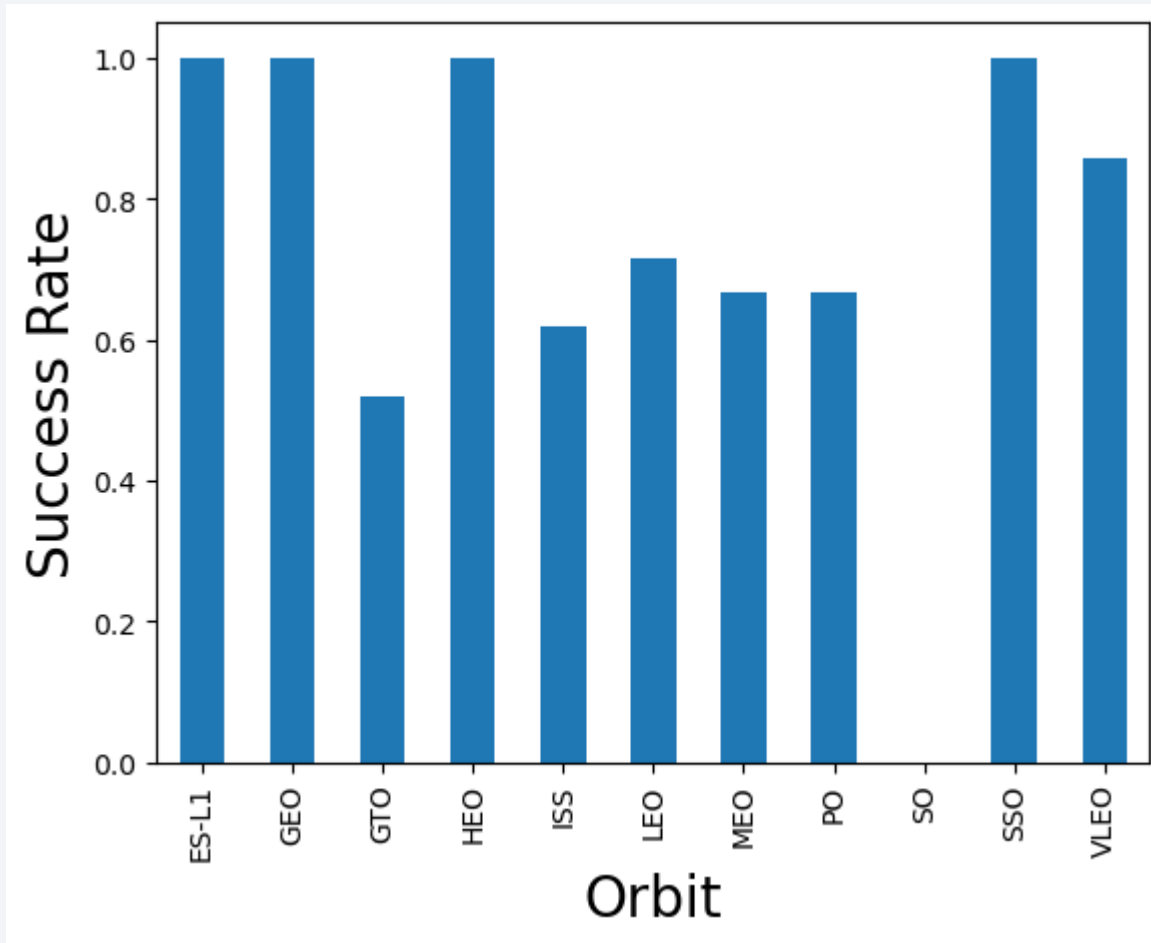
- Scatter plot compares launch sequence against launch site, colored by landing outcome.
- Later flights show more successful landings, indicating operational improvement over time

Payload vs. Launch Site



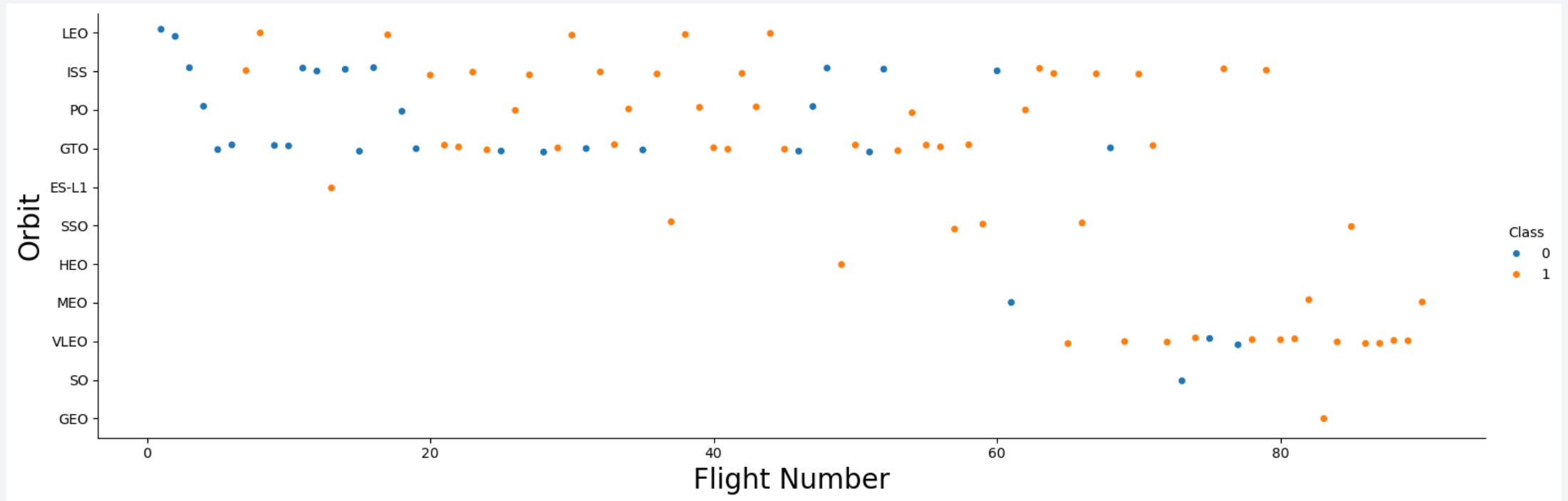
- Scatter plot compares payload mass across launch sites and landing outcomes.
- Payload mass varies substantially by launch site.
- This view helps assess whether heavier payloads are associated with lower landing success.

Success Rate vs. Orbit Type



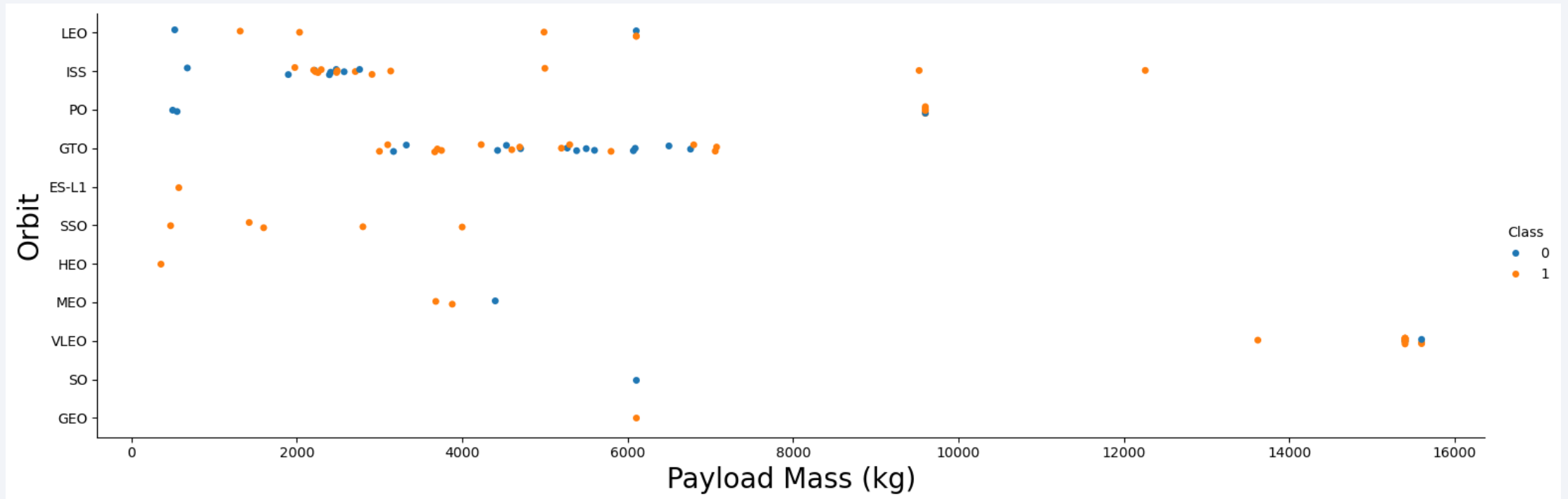
- The bar chart compares average landing success across orbit categories.
- The purpose is to identify orbit types associated with higher or lower first-stage recovery rates.

Flight Number vs. Orbit Type



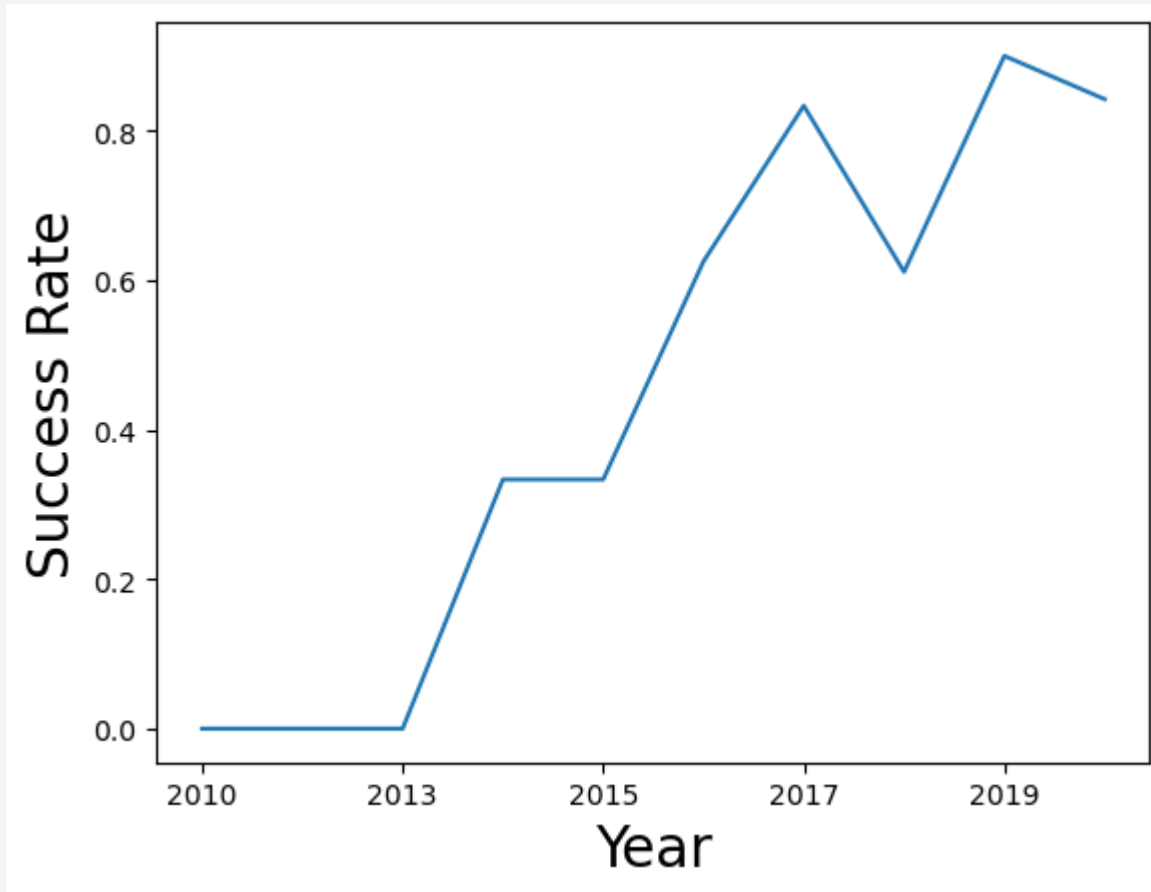
- Scatter plot evaluates how launch sequence relates to orbit type and landing success.
- Purpose: detect whether success improved over time within specific mission/orbit categories.
- Finding: later flights generally show stronger recovery performance, supporting the learning-curve hypothesis.

Payload vs. Orbit Type



- Scatter plot compares payload mass by orbit type and landing outcome.
- Purpose: identify whether certain orbits involve heavier payloads and different recovery difficulty.
- Finding: payload mass and orbit are important mission features for modeling landing success.

Launch Success Yearly Trend



- The line chart tracks the average landing success rate by launch year.
- Purpose: evaluate whether SpaceX recovery performance improved over time.
- Finding: the overall trend suggests higher success rates in later years as booster recovery matured.

All Launch Site Names

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

The query identified four unique launch sites in the mission dataset.

SQL notebook:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/04-eda-sql.ipynb>

Launch Site Names Begin with 'CCA'

Date	Booster	Launch Site	Outcome
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	No attempt

Filtered records where launch site begins with CCA and returned the first five matching launches.

SQL notebook:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/04-eda-sql.ipynb>

Total Payload Mass

Customer	Total Payload Mass (kg)
NASA (CRS)	45,596

NASA (CRS) missions carried a combined payload mass of 45,596 kg.

SQL notebook:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/04-eda-sql.ipynb>

Average Payload Mass by F9 v1.1

Booster Version Filter	Average Payload Mass (kg)
F9 v1.1	2,534.67

The average payload mass for booster version F9 v1.1 was approximately 2,534.67 kg.

SQL notebook:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/04-eda-sql.ipynb>

First Successful Ground Landing Date

Landing Outcome	First Date
Success (ground pad)	2015-12-22

The first successful ground-pad landing occurred on 2015-12-22.

SQL notebook:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/04-eda-sql.ipynb>

Successful Drone Ship Landing with Payload between 4000 and 6000

Booster Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

These boosters successfully landed on a drone ship with payload mass between 4,000 and 6,000 kg.

SQL notebook:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/04-eda-sql.ipynb>

Total Number of Successful and Failure Mission Outcomes

Mission Outcome	Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Most mission outcomes were successful; only one mission failed in flight in this dataset.

SQL notebook:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/04-eda-sql.ipynb>

Boosters Carried Maximum Payload

Booster Versions with Max Payload

F9 B5 B1048.4,
F9 B5 B1049.4

F9 B5 B1051.3,
F9 B5 B1056.4

F9 B5 B1048.5,
F9 B5 B1051.4

F9 B5 B1049.5,
F9 B5 B1060.2

F9 B5 B1058.3,
F9 B5 B1051.6

F9 B5 B1060.3,
F9 B5 B1049.7

Multiple Falcon 9 Block 5 boosters carried the maximum payload mass.

SQL notebook:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/04-eda-sql.ipynb>

2015 Launch Records

Month	Booster Version	Launch Site	Failures
01	F9 v1.1 B1012	CCAFS LC-40	1
04	F9 v1.1 B1015	CCAFS LC-40	1

In 2015, failed landing outcomes in this filtered result occurred in January and April at CCAFS LC-40.

SQL notebook:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/04-eda-sql.ipynb>

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Landing Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

The most frequent outcome in the selected period was “No attempt,” followed by drone-ship successes and failures.

SQL notebook:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/04-eda-sql.ipynb>

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

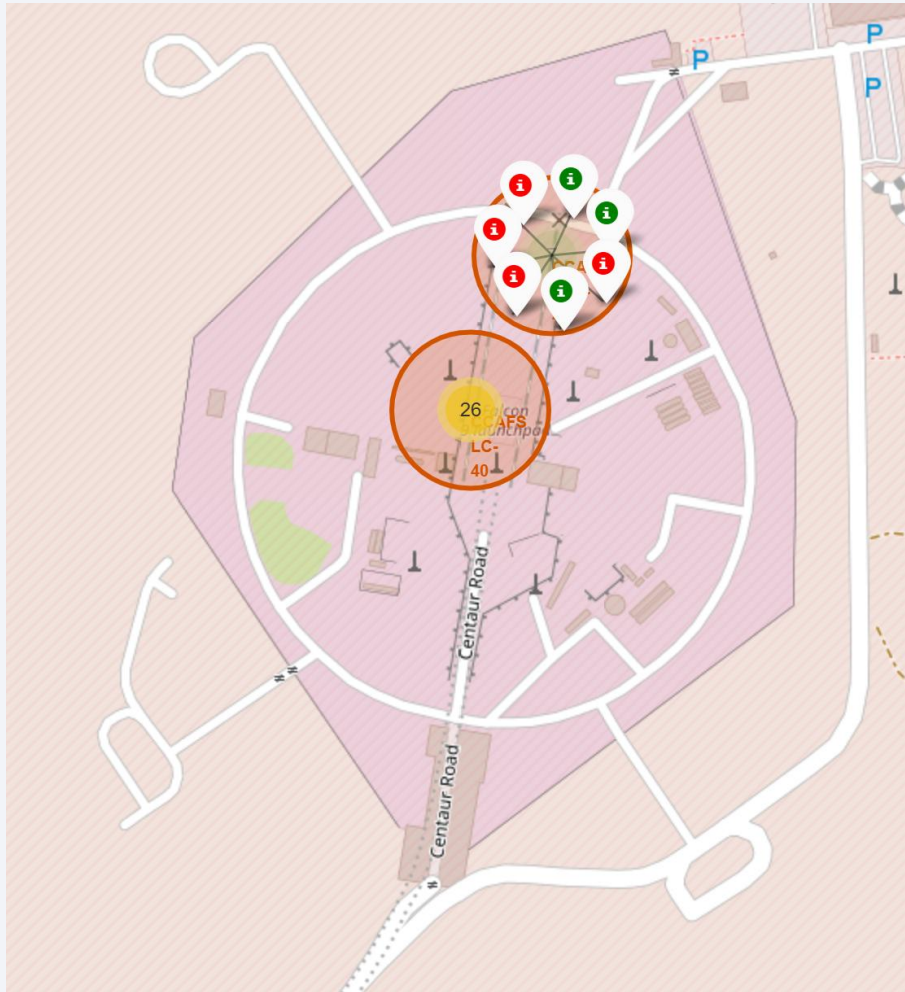
Launch Sites Proximities Analysis

Launch Site Location Markers



- The map helps verify the geographic distribution of launch infrastructure.
- We can see that all launch sites are near the coast with one site in the west and three launch sites in the west coast

Launch Outcomes by Site

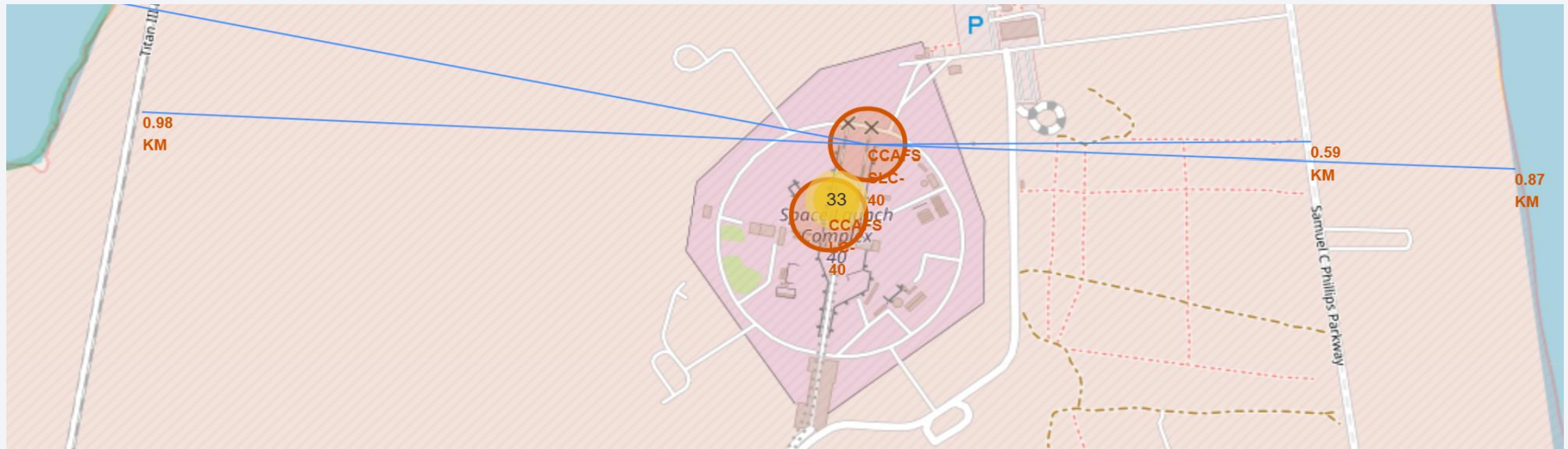


- Color-labeled markers summarize success and failure outcomes by launch site.
- Marker clustering makes multiple records easier to explore interactively.
- This helps compare landing performance geographically.

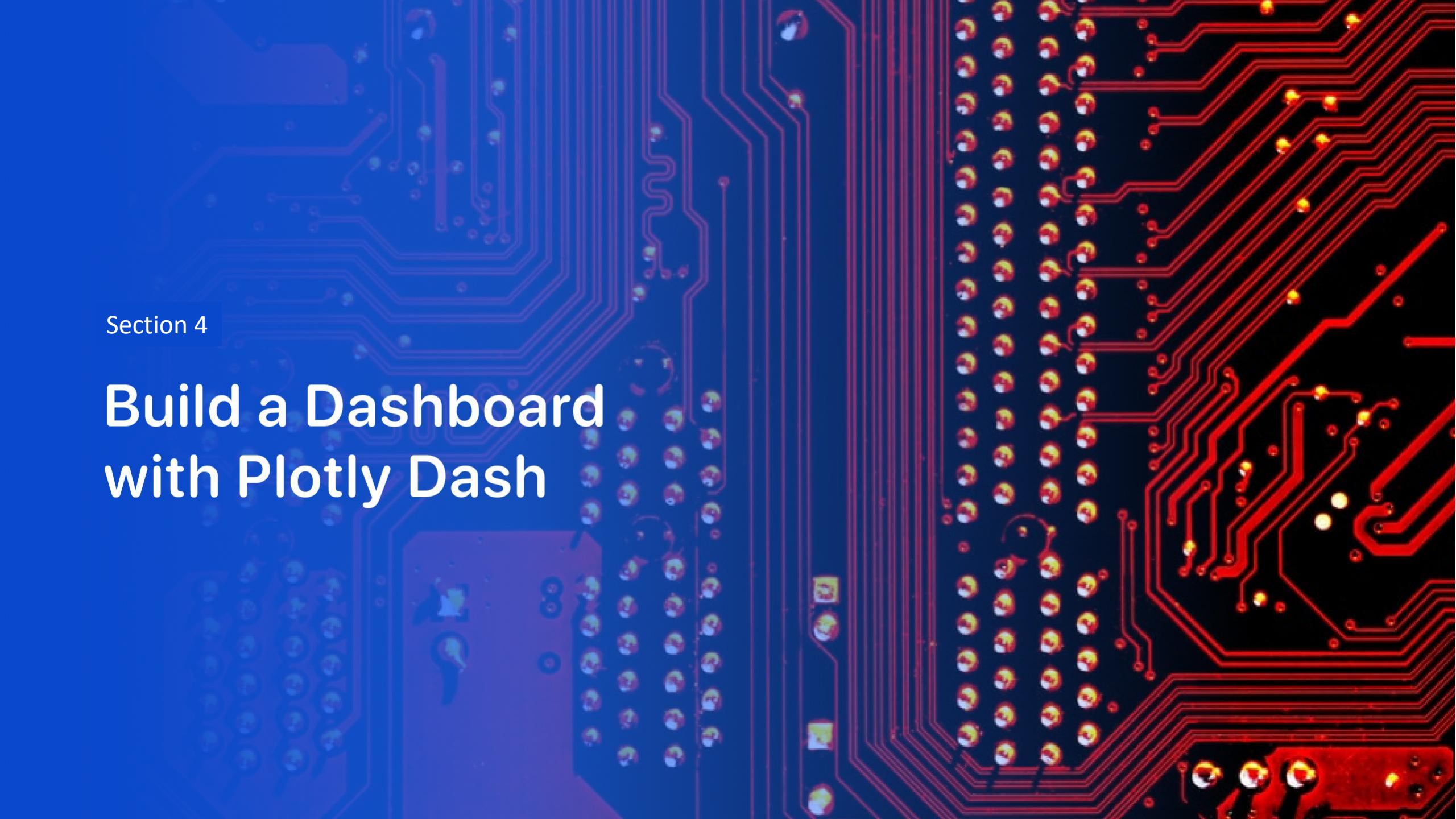
GitHub URL:

<https://github.com/lgerardoreyes14/testrepo/blob/main/notebooks/spacex/06-launch-site-map-folium.ipynb>

Launch Site Proximity Analysis



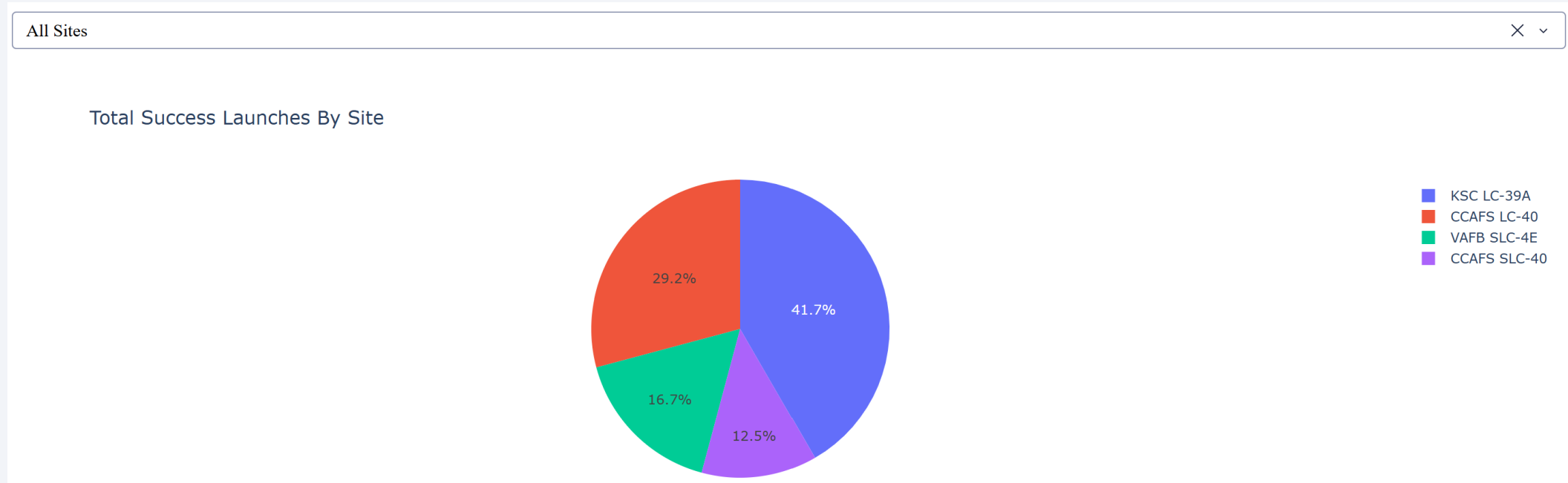
- Distance markers and lines were added to nearby coastlines, highways and railways.
- Purpose: evaluate whether launch sites are near useful geographic and transport features.
- This supports reasoning about why launch sites are located near coastal and infrastructure areas.



Section 4

Build a Dashboard with Plotly Dash

Dashboard: Successful Launches by Site



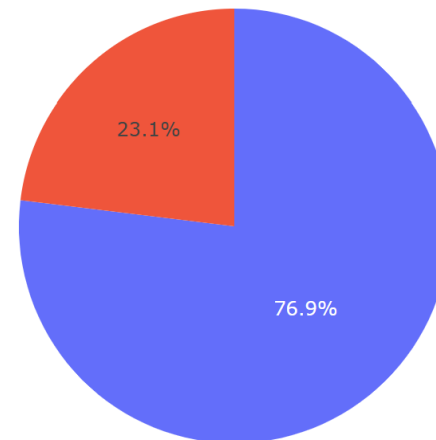
- Pie chart summarizes successful launches across all sites.
- The all-sites view supports quick comparison of site contribution to total successes.
- Interaction: site dropdown controls whether results show all sites or one selected site.

Dashboard: Highest Success Ratio Site (KSC LC-39A)

KSC LC-39A



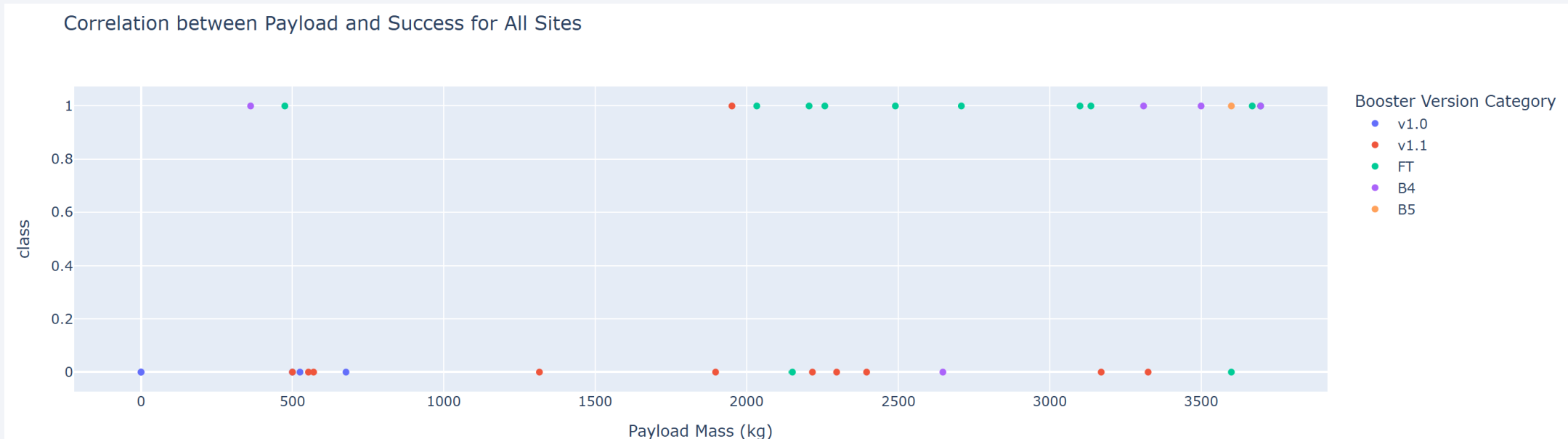
Total Success Launches for site KSC LC-39A



■ 1
■ 0

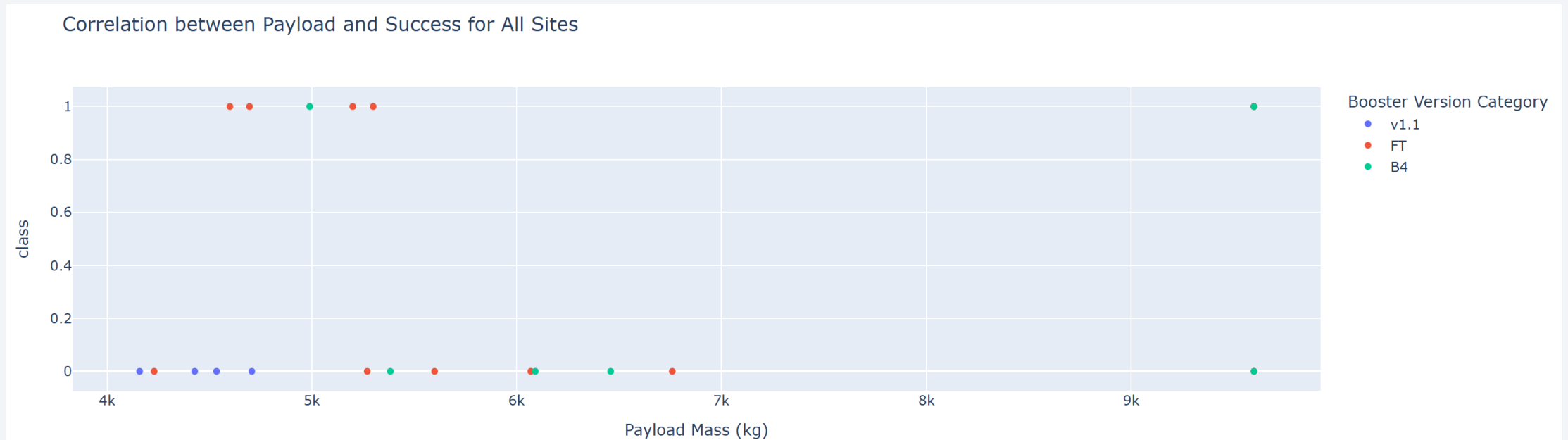
- Selected-site pie chart compares success versus failure outcomes.
- This view identifies the launch site with the strongest success ratio in the dashboard dataset.
- Interaction: choosing a site s rate is 76.9% for KSC LC-39A launch site.

Dashboard: Payload vs. Launch Outcome



- The scatter plot shows payload mass versus launch outcome for the 0–4 ton payload range.
- Booster version category is encoded by color to help identify performance differences across rocket versions.
- Within this payload range, landing success appears to increase as payload mass rises, suggesting a positive relationship between payload mass and launch outcome.

Dashboard: Payload vs. Launch Outcome



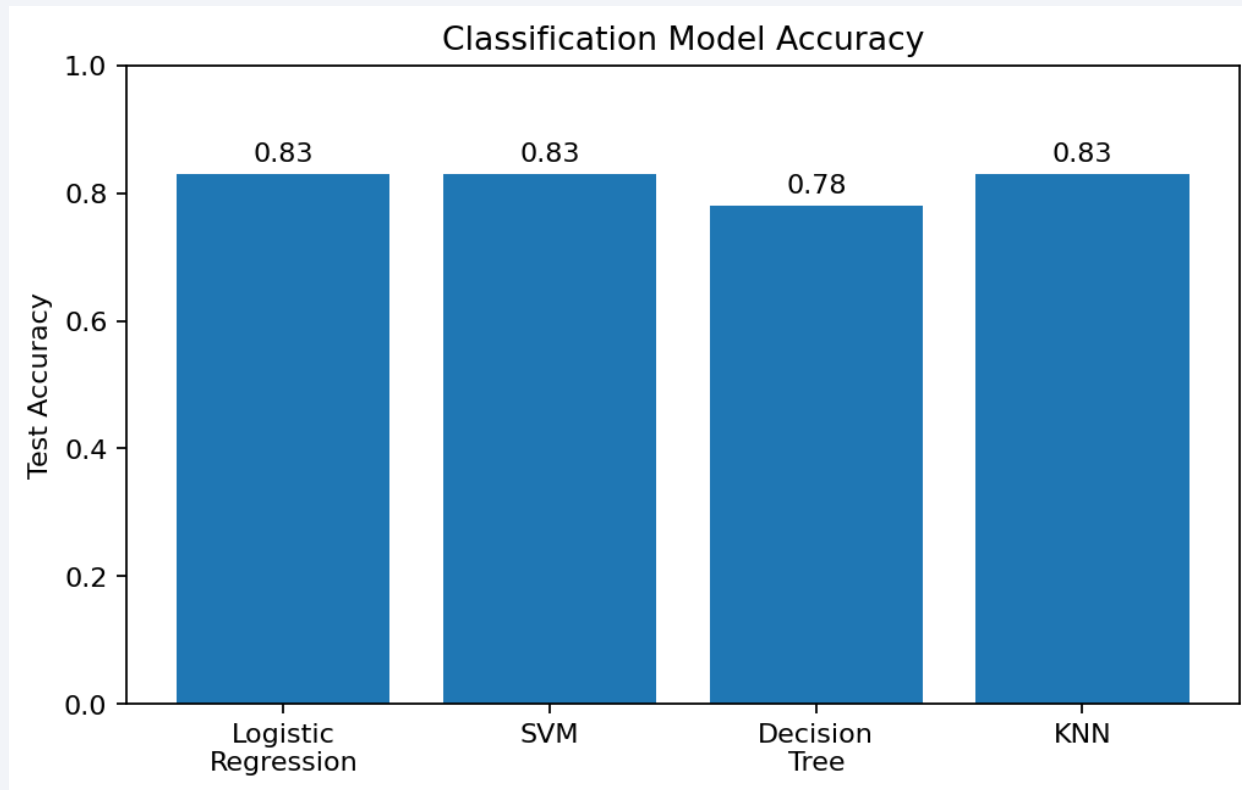
- The scatter plot shows payload mass versus launch outcome for the 4-10 ton payload range.
- Booster version category is encoded by color to help identify performance differences across rocket versions.
- Only 3 out of the 5 boosters can manage payloads higher than 4 ton
- Within this payload range, landing success appears to be maximized when the payload is about 5 ton.



Section 5

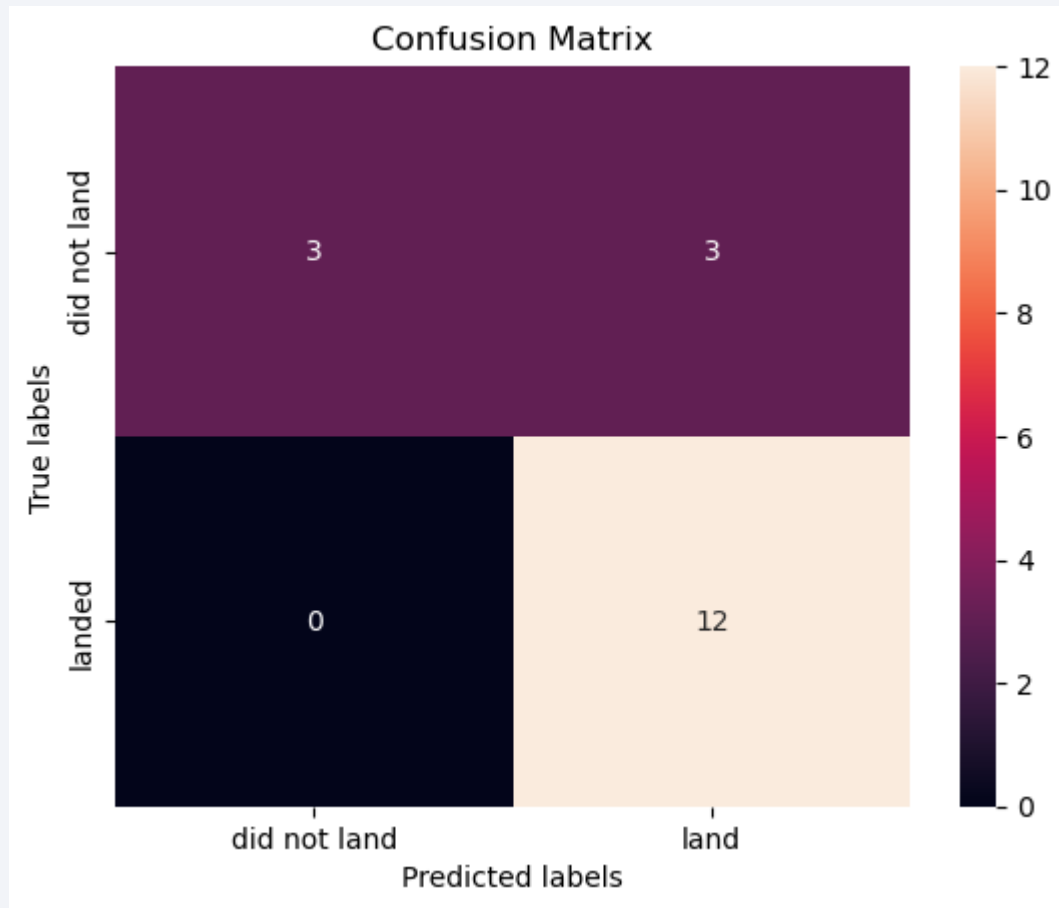
Predictive Analysis (Classification)

Classification Accuracy



- Multiple classifiers were tuned using GridSearchCV and evaluated on test data.
- Models compared: Logistic Regression, SVM, Decision Tree, and KNN.
- Most model performed pretty well, achieving 83% accuracy on the test set.
- Decision Tree model showed the lowest accuracy value (78%)

SVM Confusion Matrix



- Confusion matrix compares predicted landing outcomes against true outcomes.
- Interestingly the model does not have any false negative.
- All the errors are false positives (3) meaning that the model predicted 3 times that the booster would land when it didn't land

Conclusions

- SpaceX landing success improved over time, reflecting operational learning and booster recovery maturity.
- Launch site, payload mass, orbit, booster version, and mission characteristics are useful features for understanding landing outcomes.
- SQL and EDA results show that mission outcomes and landing outcomes vary across sites, boosters, payload ranges, and time periods.
- Folium maps and Dash dashboard views make the analysis easier to explore interactively and support spatial and operational insights.
- Classification models can support estimation of first-stage landing success, which is directly tied to launch cost competitiveness.

Appendix

- GitHub repository: <https://github.com/lgerardoreyes14/testrepo>
- EDA visualization notebook: `notebooks/spacex/05-eda-visualization.ipynb`
- EDA SQL notebook: `notebooks/spacex/04-eda-sql.ipynb`
- Folium notebook: `notebooks/spacex/06-launch-site-map-folium.ipynb`
- Dash app: `apps/spacex_dash/app.py`
- Predictive analysis notebook: `notebooks/spacex/07-machine-learning-prediction.ipynb`
- Processed data: `data/processed/spacex_launch_dash.csv` and `data/processed/spacex_launch_geo.csv`

Thank you!

